Angular responsivity models of the photodiode (Thorlabs PDA100A2)

Kevin Acuña-Condori

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The received signal strength of the single-LED system is

$$\mathrm{RSS} = A\,R(\psi)\,\cos^{m}\phi\,/\,d^{2}, \qquad \cos\psi = \mathbf{n}^{\mathsf{T}}\mathbf{u}, \tag{1}$$

where ψ is the incidence angle on the PD (normal $\mathbf{n}$, direction to the LED $\mathbf{u}$), ϕ the emission angle, d the distance, and $R(\psi)$ the normalized angular responsivity of the PD ($R(0) = 1$). Three models of R were compared; the values below are the ones currently used (`demo/config/demo.toml`, implementation in `demo/vlp/responses.py`).

1. SQ model (Bastiaens et al., IEEE Access 2020) — used

$$R_{\mathrm{SQ}}(\psi) = \max\!\left(0,\ 1 - a\,\psi^{2}\right), \qquad a = \frac{1 - 1/\sqrt{2}}{\psi_{3\mathrm{dB}}^{2}}. \tag{2}$$

It is quadratic in the angle itself (not in its cosine), and $R_{\mathrm{SQ}}(\psi_{3\mathrm{dB}}) = 1/\sqrt{2}$. It has a natural zero at $\psi_0 = 1/\sqrt{a}$, so no separate field of view is needed.

2. Generalized Lambertian model

$$R_{\cos^{m_R}}(\psi) = \begin{cases} \cos^{m_R}\psi, & \psi \le \psi_{\mathrm{FOV}} \\ 0, & \psi > \psi_{\mathrm{FOV}} \end{cases} \tag{3}$$

3. Lambertian model

$$R_{\cos}(\psi) = \begin{cases} \cos\psi, & \psi \le \psi_{\mathrm{FOV}} \\ 0, & \psi > \psi_{\mathrm{FOV}} \end{cases} \tag{4}$$

Model	Parameter	Value	Source
SQ	$\psi_{3\mathrm{dB}}$	0.61 rad (34.95°)	Bastiaens 2020, Table 1 (PDA100A2)
	a	$0.7871\,\mathrm{rad}^{-2}$	from $\psi_{3\mathrm{dB}}$
	zero $\psi_0 = 1/\sqrt{a}$	1.127 rad (64.58°)	from $\psi_{3\mathrm{dB}}$
$\cos^{m_R}$	m_R	1.9	Bastiaens 2020, Table 1 (PDA100A2)
	ψ_{FOV}	64°	hard cut-off (close to the SQ zero)
cos	ψ_{FOV}	64°	hard cut-off

Table 1: Parameters currently used.

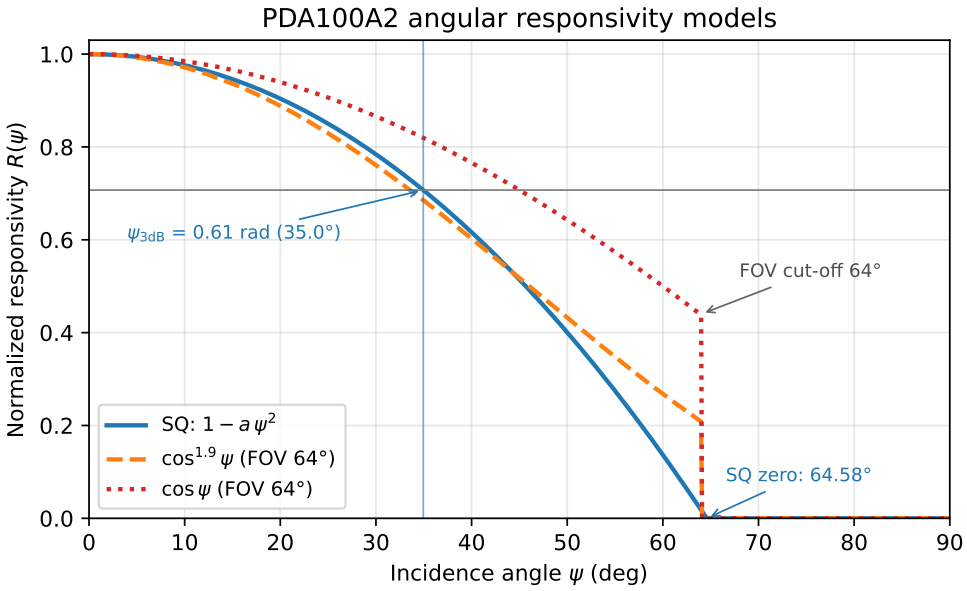


Figure 1: The three models with the values of the table. The horizontal line marks $1/\sqrt{2}$.

Effect on 3D positioning (experiment 01)

On the 125 positions of the two $K = 10$ campaigns (NLS estimator), with A and m refitted for each PD model from the same horizontal scan:

	SQ	$\cos^{1.9}\psi$	$\cos\psi$
Mean 3D error	**3.92 cm**	8.42 cm	77.4 cm
Fitted Lambertian order m of the LED	6.15	5.89	6.79

Shape test. The shape of the K RSS, independent of A and m, gives an rms deviation of 1.76 % for SQ, 1.91 % for $\cos^{1.9}$ and 7.15 % for cos (nominal geometry). **Adapter tilt.** The adapter tilt inferred from those shapes is 14.8° (SQ, corrected Z scale), 14.5° ($\cos^{1.9}$) and 24.9° (cos), against a physical value of 15°. **Conclusion.** The Lambertian model is clearly rejected. SQ and $\cos^{1.9}$ fit the shapes similarly, but SQ halves the 3D error. These results use the LED at $(270, 890)$ mm.